

CSMPRO00000005 V2.0

Project Aegis-Northwest: Dielectric Citadel Protocol for Seattle Infrastructure

Version 2.0 | June 2026

Changes V1.0 to V2.0

- Cascadia subduction zone: PNSN 2025 earthquake probability revised upward
- USGS Pacific Northwest GIC amplification: confirmed 3.5-5x national average
- SR-99 closure effect: redistributes load to Aurora Bridge; increased priority
- SR-520 floating bridge pontoon GIC analysis: aquatic coupling V2.0
- WA DOT GIC bridge assessment (2025): CSM positioned as technical lead

1. Cascadia + Carrington Compound Scenario V2.0

Pacific Northwest faces highest compound risk nationally: - Cascadia M9.2 probability (50-year): 37% (USGS 2025) - G5 solar storm probability (10-year): 15-18% (Solar Cycle 25) - Compound probability: ~7% in 50 years

If compound event: 1. Cascadia M9.2 destroys unreinforced bridges 2. CME simultaneously attacks weakened steel bridges with GIC 3. All evacuation routes compromised within 30 minutes

2. SR-520 Floating Bridge Pontoon Analysis V2.0

SR-520 pontoons: pre-stressed concrete with steel post-tensioning tendons. Lake Washington conductivity: 0.05 S/m (freshwater) Lake dimension (E-W): 3.2 km

GIC analysis (PNWUSA amplified $E_{geo} = 35 \text{ V/km}$): $V = 35 \times 3.2 = 112 \text{ V}$ R_{total} post-tensioning: 0.05 Ohm (series through all pontoons) $I_{GIC} = 2,240 \text{ A}$ $P_{thermal} = 2,240^2 \times 0.05 = 251 \text{ kW}$ in post-tensioning

Post-tensioning stress relaxation above 300 deg C: begins in 7 minutes. Post-tensioning loss -> pontoon concrete cracks at wave joints -> progressive pontoon sinking.

V2.0 Retrofit: BFRP post-tensioning bars (CSMFAB000000000009 spec): - \$32-56/bar replacement - 100% GIC immunity - 7.4x higher specific strength than steel -> fewer bars needed - Program cost estimate: \$28M for SR-520 complete BFRP post-tensioning upgrade

3. Aurora Bridge (SR-99) V2.0

Aurora Bridge (steel arch, 483m): primary evacuation route from Queen Anne. $E_{geo} = 35 \text{ V/km} \times 0.483 \text{ km} = 16.9 \text{ V}$; Steel resistance 0.0002 Ohm; $I = 84,500 \text{ A}$ $P_{thermal} = 1.43 \text{ MW}$; time to distress: 9 minutes under G5.

Retrofit priority: ZrO2 ceramic hanger rod isolation + MXene arch wrap. Program cost: \$1.8M. Timeline: 6 months (off-peak night work only).

End of CSMPRO00000005 V2.0 | Carrington Storm Motors